

Sensors

There are many different types of sensors that can be classified according to their area of application.

For creating a project, you need sensors to sense the environment and actuators to perform the required action.

Project ideas

In this section, we'll look at some possible projects using the Arduino™ platform.

Automatic watering system for gardens

The automatic garden watering system measures the moisture content in the soil to check if the plants need to be watered.

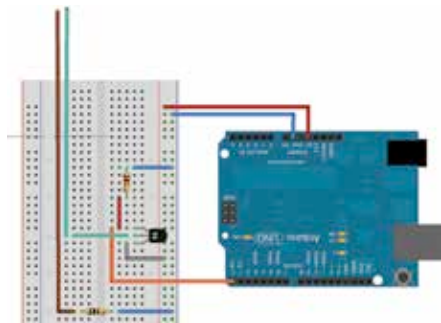
The watering system works on the concept of conductivity of water increasing with increase in moisture content or, taking it the other way round, the resistivity of soil decreasing with increase in the soil's moisture content. This can be measured using the circuit shown in the figure.

To make an automatic watering system for a garden, you'll need the following components:

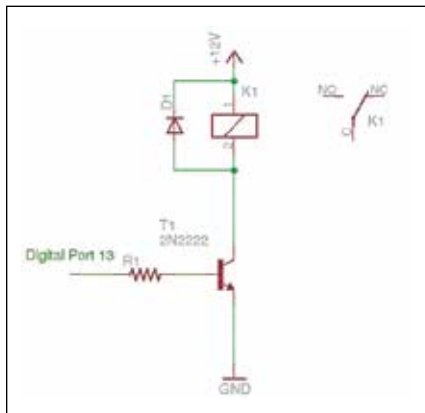
- ◆ Arduino™ board
- ◆ Wires
- ◆ Resistor
- ◆ Relay
- ◆ Transistor 2N2222

How to implement:

1. Connect the Arduino™ board to measure the resistance of soil.
2. Measure the voltage



Circuit for measuring resistance of soil



Relay circuit for using motor